

An approach and preliminary model of integrating ecological and economic constraints of environmental quality in the Guayas River estuary, Ecuador

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Abstract

We describe conceptual and simulation models of land use within the intertidal zone of the Guayas River estuary to quantify the contribution of mangrove wetlands to maintaining environmental quality of a tropical estuary. The goal of this exercise is to demonstrate the important consideration of ecological constraints in determining economic and management decisions; and how modeling can be used to quantify impacts of land use such as loss of mangrove wetlands on environmental quality. Our conceptual model treats solar energy, river flow, and tides as forcing functions that control the properties of estuarine ecosystems, but also describes market forces and cultural policies as constraints on properties of socioeconomic systems. The controversy of coastal resource management in Ecuador centers around the relative impacts of shrimp pond construction and management as negative feedbacks to the environmental quality of the Guayas River estuary. Unique oceanographic processes and land use changes contribute to complex issues of water and habitat quality in this tropical estuary, the largest estuarine complex on the Pacific coast of South America. A dynamic box model was developed for the estuary and calibrated with data collected from a 14 month survey of water quality parameters throughout the estuary. Scenarios included conversion of mangroves to shrimp ponds in three regions of the estuary, and the construction of a dam by varying three different rates of river discharge at 100, 50 and 10% of 1989 base flow. Good water quality is maintained by the low residence time of water in the estuary (11 d) because of seasonally high river flow and tidal exchange. With a 90% reduction of mangrove forests in the estuary caused by shrimp pond construction, total nitrogen concentrations increased 5 fold. However, as river discharge decreased to 10%, the same construction caused a 60 fold increase in nitrogen concentrations to 250 μM . Increases in nitrogen concentrations were higher in the upper estuary region, with much less change in the lower estuary. Thus the sensitivity of environmental quality to changes in land use in the intertidal and upland zone are linked to the hydrography of the estuary and is site specific. In the future, the combinations of these ecological models together with economic analyses of the goods and services of mangroves may provide better techniques to evaluate the economic impacts of specific coastal zone management decisions. © 1998 Elsevier Science Ltd. All rights reserved.

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1. Introduction

Estuaries have diverse ecological functions that provide goods and services to regional and global economies including commercial (both mariculture and

trawl operations) and sport fisheries (Costanza et al., 1997). Conflicts of sustaining these economic fisheries demonstrate how land use decisions need to include the direct and indirect values of natural resources such as wetlands (Farber and Costanza, 1987; Costanza et al., 1989; Dixon, 1989). In many coastal regions throughout the tropics, land use decisions in estuaries

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are favoring the expansion of shrimp mariculture, usually at the expense of natural resources, such as mangroves. Yet, these decisions fail to include the opportunity costs of their potential impacts on the habitat and water quality functions of estuarine ecosystems by improper land use of the intertidal and adjacent landscape. In addition, opportunity costs associated with the environmental quality of estuaries have to include the river basin to include many complex interactions of land use activities in the watershed (Gottfried et al., 1996). The goods and services of tropical estuaries depend on diverse habitats such as seagrasses, mangroves, benthos, and planktonic communities, which are located within the immediate boundary of the coastal zone. The environmental quality of these communities is also influenced by upland watersheds and offshore coastal processes that are coupled by rivers and tides, respectively, to the estuary. Specific types of land use practices over large spatial scales can have significant impacts on the environmental quality of an estuary. And the response of estuaries to these landscape scale changes in the tropics can vary depending on climate, geomorphology and oceanographic conditions of a region. Valuation methods must be able to account for the relative constraints of ecological and economic processes on sustainable management at different configurations of

land use across diverse coastal landforms (Repetto, 1992). These methods must be able to identify the negative and positive feedbacks of land use decisions across appropriately defined regional scales that encompass the unique value of estuarine ecosystems.

The properties of estuarine ecosystems are determined by geomorphology and geophysical characteristics of the coastal zone, including river discharge from upland watersheds (with nutrients and suspended sediments) and processes of the coastal ocean (Fig. 1; see Day et al., 1989). Groups of specific types of estuarine ecosystems can be classified into environmental settings depending on the balance of river discharge, tides, winds, and regional geomorphology (Odum and Copeland, 1972; Mann, 1975; Welsh et al., 1982; Thom, 1982, 1984; Kjerfve, 1989). Ecosystem properties include the structure, function, and relative spatial distribution of pelagic, benthic, and intertidal communities (including mangroves) of an estuary. The diversity and coupling of processes among these habitats determines the patterns of productivity, trophic food webs, and nutrient cycles that are characteristic of estuaries in a specific type of environmental setting. For instance, the structure and function of mangroves have been associated with specific types of environmental settings ranging from deltas to carbonate platforms (see reviews by Twilley et al., 1996; Twilley,

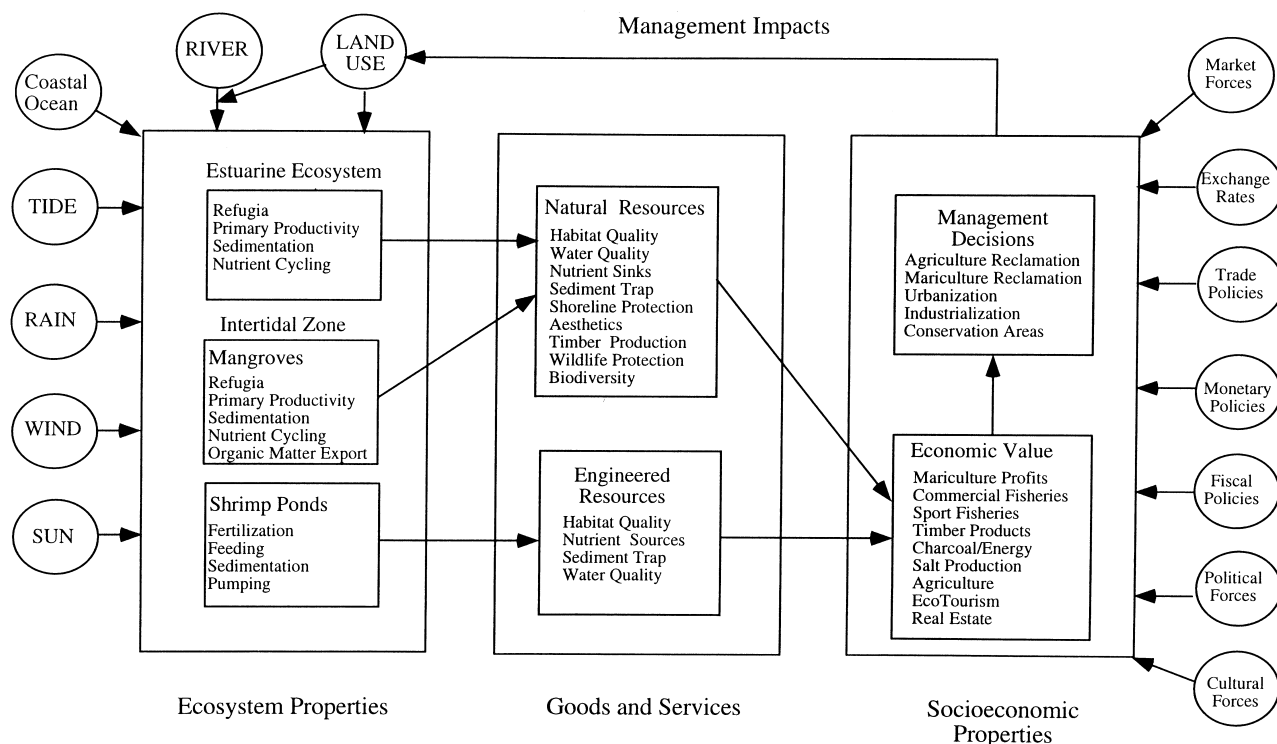


Fig. 1. Conceptual framework for the forcing functions of ecological and economic processes that determine management decisions in coastal environments.

1997a). Significance of ecological processes such as primary productivity, refugia, nutrient cycling, sedimentation, and detritus export can be associated with a particular type of environmental setting, which allows generic evaluations of specific site conditions. Tides couple the ecological processes of the intertidal zone such as mangroves with structure and function of pelagic and benthic communities within tropical estuaries. As a result of these environmental constraints and ecological processes, mangroves provide goods and services (functions) of interest to humans including habitat quality, water quality, shoreline protection, aesthetics and biodiversity (Twilley, 1997a,b; Ewel et al., 1998). These functions can contribute either directly or indirectly to the economic value of the estuary and surrounding landscape.

Shrimp ponds are artificial ecosystems located in the estuarine drainage area, either within the intertidal zone, thereby replacing mangroves, or just inland of intertidal vegetation. The ecosystem properties of shrimp ponds are controlled in part by the environmental quality of the estuary, since water is pumped from the estuary to the ponds, and also by subsidies in nutrients and feed provided by management operations. Good water quality improves the yield of shrimp and influences the intensity of pond operations. Also, since waters of shrimp ponds are returned back to the estuary, ecological processes of ponds (as influenced by pond management) will influence the environmental quality of the estuary. Ponds are managed by decisions such as rates and types of fertilization, feed inputs and water exchange, that cause these ecosystems to acquire attributes different from those of the estuary. These attributes of ponds provide goods and services of more specific value to humans such as elevated yields of monoculture shrimp communities. These goods and services are referred to as engineered resources in comparison to those services of natural resources (Fig. 1). Although the services of shrimp ponds are heavily controlled by the socioeconomic system rather than the ecological processes of an estuary; the productivity of shrimp ponds is inextricably linked to the environmental quality of the estuary (including the intertidal and upland watersheds, and offshore processes), which is constrained by the environmental setting (Fig. 1).

Our conceptual model (Fig. 1) treats solar energy, river flow, and tides as forcing functions that control the properties of estuarine ecosystems; but also includes socioeconomic constraints, such as international markets, political processes, exchange rates, and monetary and trade policies. The conceptual model argues that ecological and socioeconomic forcing functions determine the value of mangroves and shrimp ponds thereby controlling coastal zone management decisions (Fig. 1). The goods and services of

mangroves, such as habitat and water quality, receive no explicit value in the conceptual model (the same as occurs in society). Mangroves are valued indirectly in terms of their contribution to shrimp pond profits by providing goods and services at no costs to the industry. Profits associated with shrimp farming are generated by the difference between income from shrimp yields, and costs associated with pond operation. The level of shrimp production and operation costs, such as dredging, construction, pumping, fertilization, and land (authorizations), depend somewhat on the environmental quality of the estuary (Twilley, 1989). Mangroves and tides provide the shrimp industry with clean water and productive habitats that enhance wild post larvae supply and shrimp yields in ponds. With the loss of free services provided by natural resources, the costs of shrimp production increases, such as providing post larvae from hatcheries, dredging to remove sediment, and pumping to control eutrophication. This conceptual model emphasizes the potential impacts of negative feedbacks from management decisions on the natural resources that can decrease potential profits of the shrimp industry. Impacts of shrimp pond operations are linked to the dependence of this industry on sustaining the environmental quality of estuarine ecosystems (Fig. 1).

This review describes the ecological linkages between shrimp ponds and mangroves associated with the mariculture industry in Ecuador to illustrate the importance of environmental quality to sustainable coastal zone management. The value of shrimp production from ponds increased from \$56.9 million US dollars in 1980 to \$482 million US dollars in 1991, ranked second only to petroleum as an export commodity for Ecuador (Olsen and Arriaga, 1989; Aiken, 1990). During early 1990's, cash generated by mariculture activity was more important to the economy of Ecuador than bananas and cacao combined, and twice as important as coffee (Aiken, 1990; Argüero and González, 1991, cited in González, 1993, p. 2). Ecuador produced less than 5,000 MT of shrimp from ponds in 1979, rising quickly to a production of more than 100,000 MT by 1991, during which time the area of ponds increased to over 100,000 ha (McPadden, 1985; Twilley et al., 1993). The controversy of coastal resource management in Ecuador centers around the relative impacts of pond construction and management as negative feedbacks to the ecological processes of mangroves and estuaries. The success of developing management plans for the coastal zone depends on the ability to identify properties that determine the environmental quality of coastal ecosystems in Ecuador, and how shrimp pond management influences the contribution of services from natural resources such as mangroves with incremental changes in land use of the intertidal zone (Fig. 1).

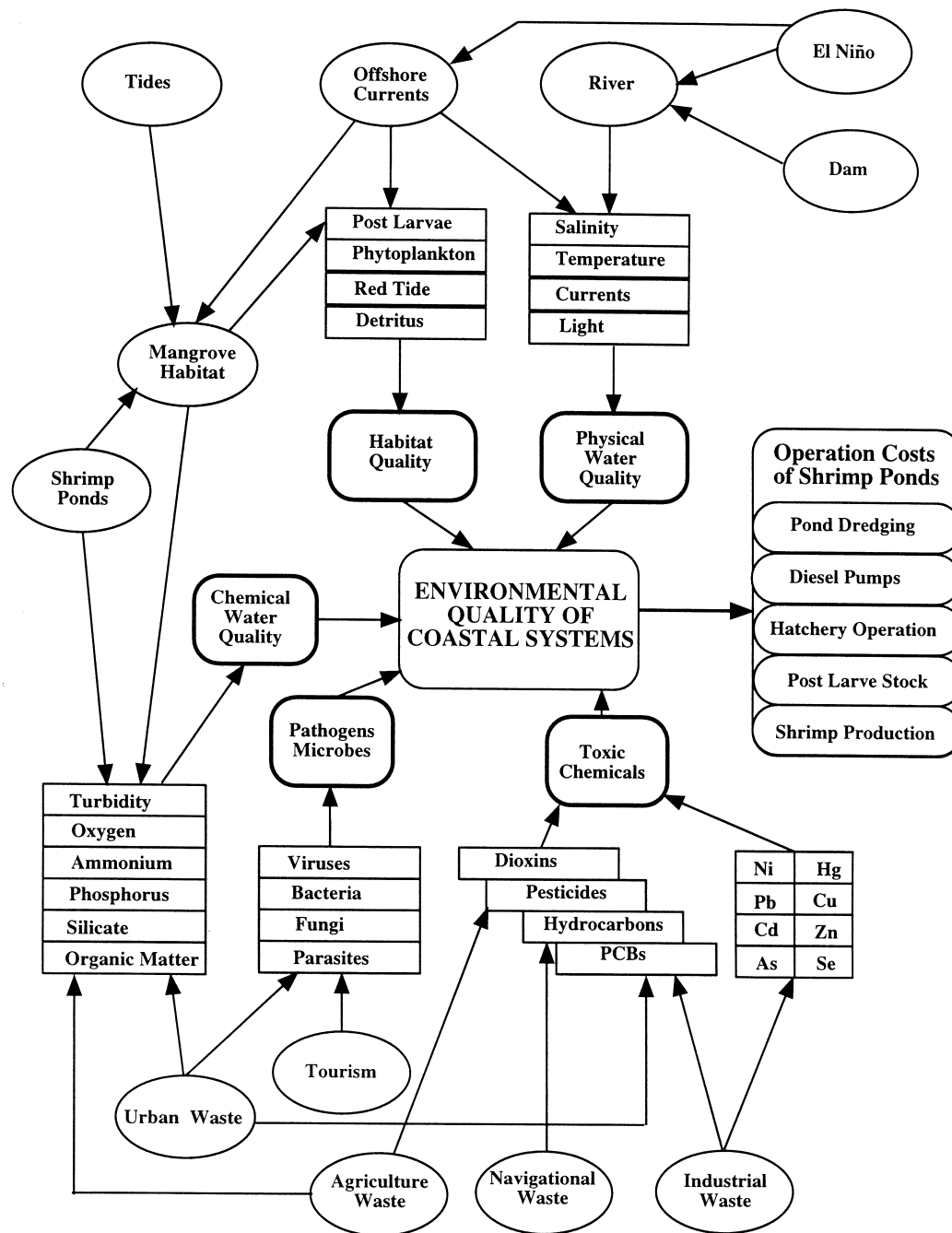


Fig. 2. The complex interactions of coastal processes (upper panels) and upland land use (lower panels) that determine the environmental quality of coastal resources in Ecuador.

2. Environmental quality of Guayas River estuary

The coastal resources of Ecuador are impacted by diverse economic activities and land use patterns that may influence the environmental quality of coastal waters (Fig. 2). Environmental quality of the Guayas River estuary is influenced by inputs from upland watersheds, exchanges with the intertidal zone, and oceanographic processes in the Gulf of Guayaquil. Activities in the watershed include a dam project that

will influence fresh water discharge, expanding agriculture with associated input of chemicals including nutrients and pesticides, sewage from increased urbanization, and toxic substances from industrial activities (Arriaga, 1989; Solórzano, 1989). In addition, red tides develop in coastal waters that may be pumped into hatcheries and shrimp ponds. These diverse anthropogenic influences on water quality in the estuary complicate environmental management of this coastal ecosystem. In the upper panel of Fig. 2,

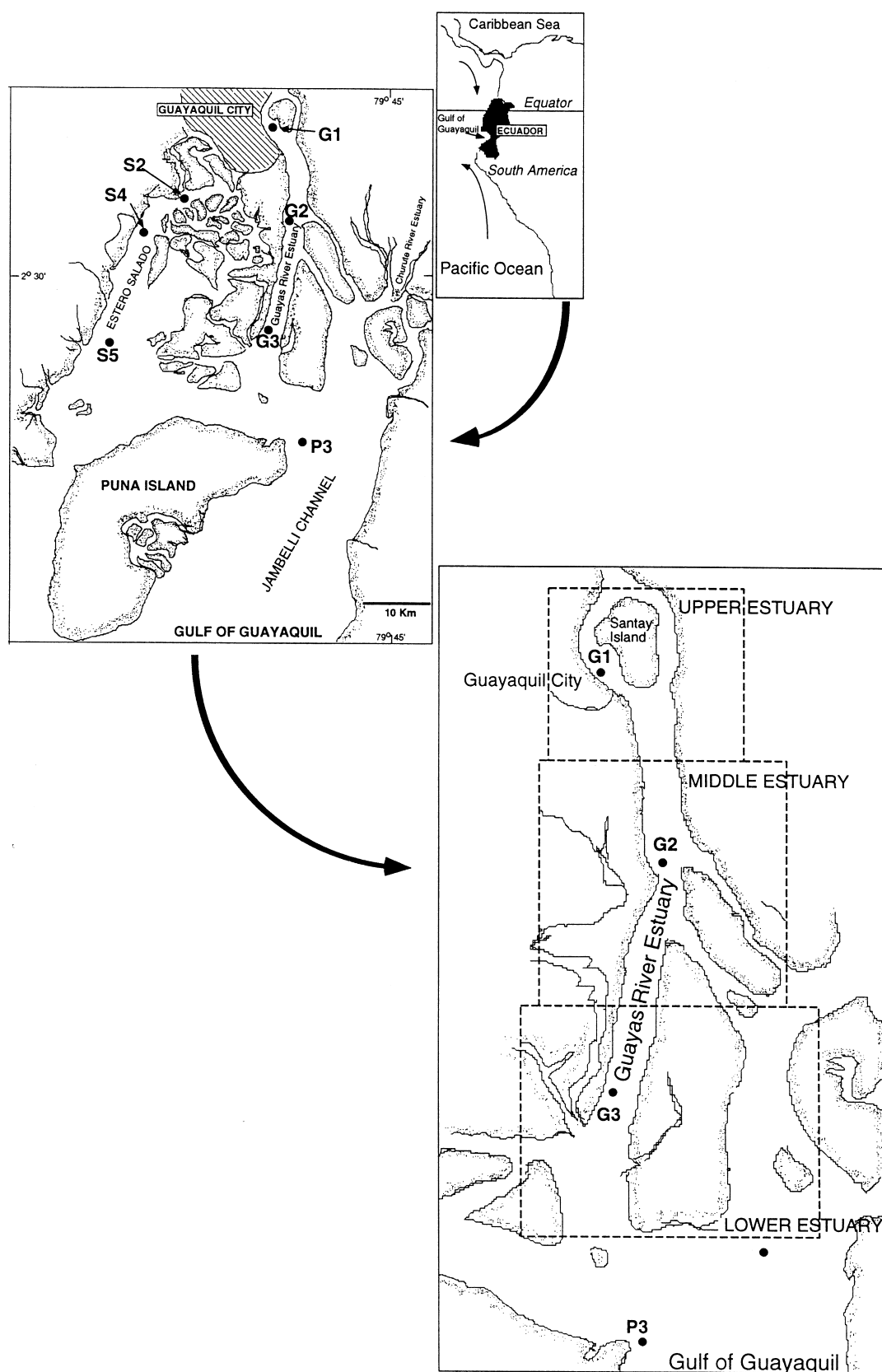


Fig. 3. Map of the coastal zone of Ecuador with details of the Guayas River estuary showing sampling stations of the 1989 water quality surveys and the boundaries of the three boxes used to simulate environmental quality of the estuary.

the environmental setting that influences the environmental quality of coastal systems is shown to include river discharge, tides, and coastal ocean currents. In the lower section of Fig. 2, more anthropogenic effects of industry, navigation, urban, agriculture, and tourist activities are linked to environmental quality.

2.1. Environmental setting

The Gulf of Guayaquil receives runoff from some 20 rivers in a watershed of 51,230 km² forming the largest estuarine ecosystem on the western Pacific coast of South America (Cucalón, 1984). The major source of fresh water is the Guayas River, which forms 60 km upstream at the confluence of the Daule and Babahoyo Rivers. The mean discharge of 1143.7 m³/s for the Guayas River is the highest among the 30 rivers in the coastal zone of Ecuador representing 39% of the total discharge from this lowland region. Mean precipitation in the Guayas River drainage system north of Guayaquil is 885 mm/yr, which may range from less than 400 to more than 1800 mm during any one year (Stevenson, 1981). Discharge is strongly seasonal ranging from 200 m³/s during the dry season to 1600 m³/s in the wet season with an average amount of precipitation (about 900 mm/yr). Tides are semi-diurnal with an equal amplitude of 1.8 m in the Gulf of Guayaquil, amplified to 3–5 m in the Guayas River estuary near the city of Guayaquil.

Coastal Ecuador is a transition zone, or equatorial front, between southerly flowing tropical water from the Panama Bight and northwardly flowing Humboldt Current from Peru (Fig. 3). Tropical water flows southward from the Panama Bight along the coast of Panama and Colombia to about 2° S supplying warm (>25°C) and low salinity (<34 g/kg) waters to the coast of Ecuador (Pesantes and Pérez, 1982). Colder (<22°C) and more saline (>35 g/kg) waters flow northward from Peru with the Humboldt current that is strongly influenced by coastal upwelling. The mixing of these two water masses occurs between Manta and Punta Santa Elena along the coast of Ecuador and gradually moves southward into the Gulf of Guayaquil. The dominance of the Panamanian Current occurs during the summer causing an increase in sea water temperature and initiates the onset of the rainy season (Cucalón, 1984, 1989). Years of abnormally warm water temperatures and high rainfall are associated with El Niño climate patterns due to the influx of unusually warm surface water in southeast Pacific Ocean. The warmer offshore waters have resulted in the explosive populations of white shrimp off the coast of Ecuador from enhanced spawning, maturation, and recruitment (Zimmerman and Minello, 1986). In the last century, 10 major El Niño events were recorded (Cucalón, 1989) and one of the

strongest is presently occurring along the coast of Ecuador. The high availability of post larvae that supported the expansion of the shrimp industry in 1983 and 1984 was associated with an unusually strong El Niño event. The unpredictable nature of oceanographic events and their influence on river discharge and near shore recruitment result in complex issues of what factors contribute to reduced availability of post larvae along the coast of Ecuador (Fig. 2).

2.2. Land use impacts

Changes in the environmental quality of the Guayas River basin and estuary are associated with land use changes both in upland and intertidal watersheds (Solórzano, 1989; Twilley, 1989). For example, it has been shown that the quality of water in the estuary may be influenced by introduction of chemicals such as nutrients and pesticides from agriculture, sewage from large urban areas, and heavy metals from industry (Fig. 2). Rivers provide the conduit that links ecological processes of an estuary with land use practices of upland watersheds (Fig. 1). Deforestation of natural vegetation followed by replacement with agroecosystems, in addition to urban and industrial activities, can change the chemical composition of riverain inputs to the Guayas River estuary. In addition, the distribution and turnover rate of these pollutants in the estuary are influenced by alterations in the quantity and seasonal nature of fresh water discharge from the watershed. Thus the quantity and quality of riverain inputs together are important linkages of the environmental quality of Guayas River estuary to the productivity of shrimp pond mariculture (Fig. 2).

A dam has been constructed at the confluence of the Daule and Peripa rivers for diversion of water supply, control of river flow, and hydroelectric power. Water is diverted with an aqueduct from the Daule River to the Santa Elena peninsula for potable water, irrigation for agriculture, and industrial use. The dam will influence the amount of water from the Daule and Peripa Rivers that normally discharges into the Guayas River. Presently the proposed operation of the dam calls for an average annual flow of from 100 to 175 m³/s (Jenkins, 1979; Arriaga, 1989). This flow will vary from a high of 321 m³/s during the wet seasons in April, to a low of 124 m³/s in August. Compared to the normal flow of the Daule and Peripa Rivers, this modified flow is much lower than the fresh water discharge of up to 1,000 m³/s that usually occurs during the wet season. During the dry season, to control salt water intrusion, the dam will provide water above the average discharge of about 50 m³/s from supplies stored in the impoundment. Based on average monthly flows, the normal discharge of 343 m³/s for these two rivers will be restricted to 174 m³/s, a reduction of

about 49%. This reduction represents a 15% loss of the fresh water to the Guayas River and 13% from the Guayas River estuary. The loss of fresh water from an estuary in a semiarid zone such as the Guayas province may influence the patterns of salinity in this coastal ecosystem.

2.2.1. Water quality

The chemical and physical characteristics of the Guayas River estuary, in the upper reaches of the Gulf of Guayaquil, were surveyed for 14 months during 1989 and 1990 (Cárdenas, 1995). The Guayas River estuary consists of three sub-estuaries: the Guayas, Churute, and Salado (Fig. 3). Salinity regimes in the Guayas and Churute sub-estuaries reflect the strong seasonal influence of fresh water with mean salinities of 18 and 16 g/kg, respectively; while fresh water input is restricted in the Salado with mean salinities of 28 g/kg (Table 1). Nitrate was the major form of fixed nitrogen present in the estuary accounting for 70% of the DIN. NO_3^- plus DON (dissolved organic nitrogen) and PN (particulate nitrogen) formed about 30% each of the total nitrogen (TN) pool. Total phosphorus concentrations in all sub-estuaries were similar with mean values ranging from 3 to 5 μM (Table 1). TN:TP ratios <9 is a distinct feature of this tropical estuary compared to a normal balance of about 16 for phytoplankton. A ratio below 16 indicates that nitrogen may be limiting productivity. However, the high turbidity of

this estuary with suspended sediment concentrations averaging from 115 to 494 mg/L in the main channel may be the key limiting factor to chlorophyll production (Table 1).

2.2.2. Mangrove resources

The most controversial issue related to the environmental quality of the coastal resources of Ecuador has been the exploitation of mangroves associated with the construction of shrimp ponds. In 1969, the total mangrove area in Ecuador was 203,695 ha, of which nearly 78% were located in the two southern provinces of Guayas and El Oro (Alvarez et al., 1989; Fig. 4C). By 1991, the total mangrove area declined to 161,410 ha, and 81% of this resource remained in Guayas and El Oro provinces (Fig. 4C). The reduction of mangrove area from 1969 to 1991 was estimated at 42,285 ha or 20.8% of the original 203,695 ha. By 1991 CLIRSEN, estimated that 145,940 ha of ponds had been constructed in the coastal zone, suggesting that 29% of shrimp ponds had eliminated mangrove areas. During this same interval, nearly 44,005 ha of salines had been lost, nearly all to shrimp pond construction. The greatest loss of mangroves has occurred in the Guayas Province at nearly 1,500 ha/yr for both the 1984–1987 and 1987–1991 periods of analysis. Because of the large area of mangroves found in the Guayas River estuary, this high rate of loss is less than 15% of the existing mangrove resources. By late 1988 the destruc-

Table 1

Mean concentration of water quality parameters in the lower and upper zones of three regions in the Guayas River estuary during the wet and dry seasons (Cárdenas, 1995)

Parameter	Units	Season	Churute		Guayas		Salado	
			lower	upper	lower	upper	lower	upper
Temperature	°C	dry	25.4	25.0	25.8	25.2	25.1	25.6
		wet	28.6	28.5	28.7	28.4	28.6	28.8
Salinity	g/kg	dry	23.0	18.0	29.7	10.5	30.3	27.2
		wet	13.2	8.0	25.4	1.6	27.6	26.0
Dissolved oxygen	mg/L	dry	3.7	3.1	4.1	4.1	4.2	3.5
		wet	3.3	2.3	5.0	3.4	4.0	2.6
Suspended sediments	mg/L	dry	332.0	95.6	315.2	455.7	207.3	93.6
		wet	242.9	58.3	114.6	494.0	121.0	48.2
Total nitrogen	μM	dry	31.7	31.8	23.2	45.6	22.8	25.3
		wet	34.7	36.8	19.2	42.7	21.2	29.7
Total phosphorus	μM	dry	4.2	4.2	5.2	5.2	4.0	5.7
		wet	3.9	4.7	3.1	5.0	3.1	3.9
PO_4	μM	dry	2.6	2.8	2.0	3.3	1.8	2.4
		wet	1.8	2.8	1.5	2.3	1.9	3.0
NH_4	μM	dry	1.8	2.8	2.1	3.8	2.6	4.8
		wet	1.8	2.3	1.5	3.8	2.0	3.2
NO_3	μM	dry	10.4	9.2	6.3	19.4	5.9	7.2
		wet	12.6	11.7	4.2	15.8	4.4	10.6
DIN^a	μM	dry	12.9	14.4	9.3	23.6	9.2	13.6
		wet	15.6	17.4	6.3	19.9	7.7	17.5
TN:TP		dry	7.6	7.6	4.5	8.9	5.7	4.4
		wet	8.9	7.8	6.2	8.5	6.8	7.6

^a DIN = dissolved inorganic nitrogen.

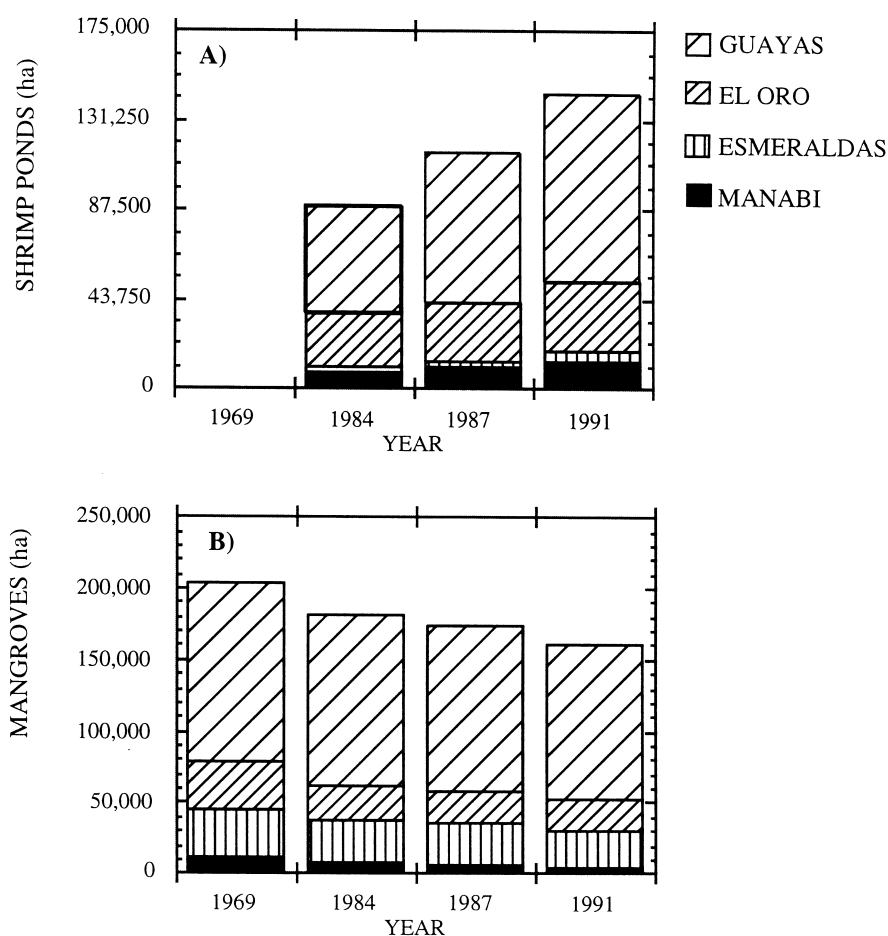


Fig. 4. Land use changes in the intertidal zone of the four coastal provinces of Ecuador from 1969 to 1991 including (A) shrimp ponds and (B) mangroves.

tion of mangrove habitat in some estuaries in the province of Manabi was virtually complete, such as in Río Chone estuary. From 1974 to 1988 mangrove area along this estuary declined from 3,973 to 600 ha, and nearly all of the mangrove loss was associated with construction of shrimp ponds. In the Atacames River estuary, there are only 50 ha of mangroves remaining of the 578 reported in 1970s, representing a loss of 90.1% of the mangrove resources. In the southern province of El Oro, the Machala–Puerto Bolívar area lost over 50% of a very productive mangrove system. The impacts of mangrove loss on the environmental quality of estuarine resources depends on specific regional land use characteristics. However, the cumulative loss of mangroves along the entire coast is also an issue relative to sustaining habitat necessary for continued recruitment of shrimp to the coastal zone (Turner, 1989).

2.2.3. Shrimp mariculture

Shrimp ponds represent managed ecosystems that are linked to the ecological processes of several coastal ecosystems (Fig. 2). The impacts of shrimp pond man-

agement on the environmental quality of estuaries is related to the loss of mangroves associated with pond construction, pond operations such as pumping, fertilization, feed supply and dredging, along with the harvesting of post larvae to stock ponds. Evaluating the interactions of shrimp pond management depends on our understanding the habitat and water quality functions of coastal ecosystems (Fig. 2). These estuarine functions are linked to a variety of ecological processes in mangroves, as well as other ecosystems in the coastal zone (Twilley et al., 1997). Specific stages of shrimp pond management depend on these free services of the estuarine ecosystem by providing a habitat to facilitate the growth of post larvae to adult shrimp. These include functions that supply wild post larvae to grow out ponds, and water quality conditions that enhance the growth and survival of juvenile shrimp. Habitat quality of mangroves in the estuary is lost when these wetlands are destroyed during the construction of ponds. Mangrove functions such as nutrient sinks that contribute to the environmental quality of the estuary are also removed. Critical water quality parameters that affect shrimp pond management, and

profits, include dissolved oxygen, turbidity, and toxic substances, including red tides (Fig. 2). Many water quality conditions in ponds, such as hypoxia, can be managed by increasing the exchange of water from the estuary that supplies more dissolved oxygen to the water column. But increased pumping also effects the environmental quality of the estuary by recirculating estuarine water that can contain elevated concentrations of nutrients. This represents a feedback effect of shrimp pond management on estuarine ecosystems referred to as 'self-pollution' (Csavas, 1994).

3. Simulation model

3.1. Model description

We constructed a simple dynamic box model of the Guayas River estuary to quantitatively test how to manage mangrove conservation with shrimp pond operations. Our model is a 'strategic simulation' model (Ross and Gurney, 1993) that focusses on realism and simplicity by selecting the ecologically relevant components and processes within the context of specific management issues described above (Ross and Gurney, 1993; Jørgensen, 1994; Hopkinson and Vallino, 1995). There exist few attempts to model aquaculture pond systems and to link them to larger ecosystems in such a way as to permit analysis of landscape management (Hagiwara and Mitsch, 1994). In shrimp mariculture most environmental studies are directed to maintaining optimal conditions in ponds for growth and development of shrimp post-larvae, and for maximizing profits (Robertson and Phillips, 1995; Dierberg and Kiattisimkul, 1996). There are few ecologically-based regional models to assess shrimp pond land use within an estuary relative to both spatial and temporal issues of sustainability (Wang, 1990). Similarly, existing models can rarely examine feedbacks of pond operations that may influence long-term economic gains of this industry by reducing water quality (e.g. self pollution). Most concerns have centered around the loss of free ecological functions provided by mangroves (Twilley et al., 1993; Twilley, 1995) and a decline in environmental quality of the estuary. We describe a model to address issues of mangrove deforestation resulting from the rapid growth of the shrimp pond industry and its potential impacts on water quality in the Guayas River estuary. The computer model was developed using the dynamic simulation software STELLA II, version 3.0.5 (Hannon and Ruth, 1994), a user-friendly object-oriented application for solution of complex differential equations.

The main objective of our ecological model is to evaluate the effects of shrimp mariculture on the transport of salt, suspended sediments, and total nitrogen

in the Guayas River estuary. These three parameters are representative of the conservative and nonconservative behavior of substances that are exchanged with water between shrimp ponds and the estuary and can modify water quality. Salt is conservative and represents changes that are associated with dilution of river discharge with sea water from the Gulf of Guayaquil. We assumed in these simulations that shrimp ponds do not alter salinity of water exchanged with the estuary, although this may not be accurate (Snedaker et al., 1986; Twilley, 1989) and needs further analyses. The behavior of salt is necessary to test our assumptions concerning the hydrography and mixing characteristics of our box model. Total suspended sediments are also strongly influenced by estuarine hydrography rather than biochemistry in the estuary, but fluxes are sensitive to land use changes in the intertidal zone between mangroves and shrimp ponds. Suspended sediment concentrations determine the quantity and quality of light in the estuary, that will influence levels of primary productivity (Cloern, 1987). Total nitrogen is influenced by estuarine hydrography, land use, along with the biochemistry of ecological processes in the Guayas River estuary, representing a more complex indicator of water quality. The model explores how the loss of mangrove wetlands along specific regions of the estuary may affect nitrogen content in the estuary, using nitrogen concentrations as an index of change in environmental quality. We consider our simulations of different management scenarios as a preliminary approach to focus future efforts to calibrate and verify other attributes of water quality, such as dissolved oxygen. Also, this 'unit' model approach can be used to build more sophisticated dynamic and landscape models in the future, whose simulations can be used to develop management plans for river basins coupled to the Guayas River estuary. This last objective is fundamental because the success of developing management plans for this region depends on the ability to integrate ecological functions of mangroves with the management of shrimp ponds (Fig. 1).

3.1.1. Hydrography

The Guayas river estuary is a partially mixed estuary with tidal current speeds up to 100 cm/s (Murray et al., 1975). The dominant upstream flux of mass and salt are apparently associated with the tidal prism (Murray et al., 1975). We divided the Guayas River estuary in three regions (upper, middle and lower, Fig. 5) to facilitate modeling the hydrology using a box model approach (Officer, 1980; Miller and McPherson, 1991). Boundaries of the system are the river in the upper region and the passage formed by the Jambeli channel (Fig. 3) in the lower region. We included water exchange between the estuary with shrimp ponds and surrounding mangroves in each of

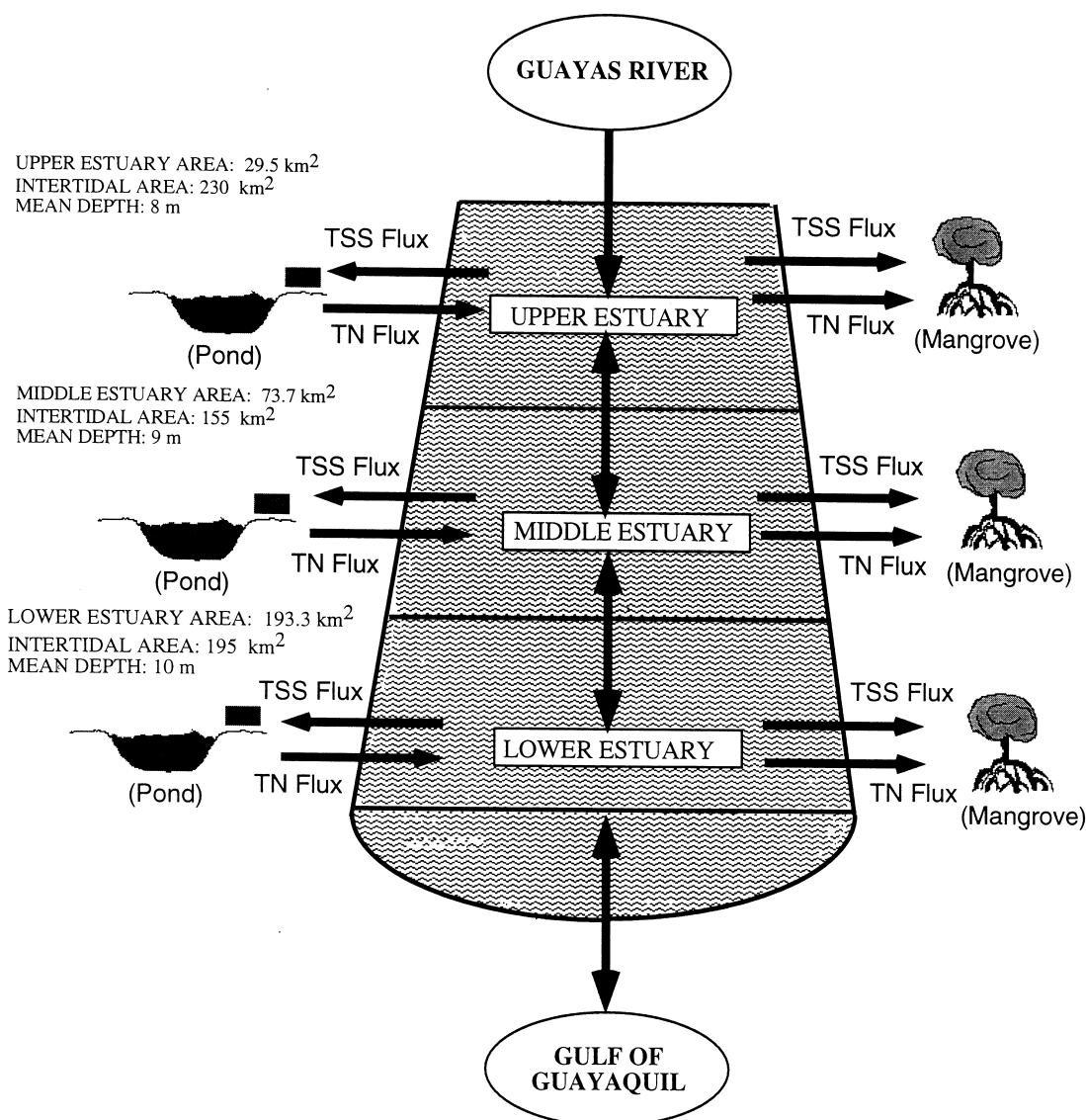


Fig. 5. Diagram describing the model used to simulate the fate of salt, sediments and total nitrogen in the Guayas River estuary based on exchanges among three boxes in the estuary, and lateral exchange of each box with corresponding region of intertidal zone.

the three regions. Salinity values in the upper and lower regions of the estuary of 8 and 20 g/kg, respectively, reflect the influence of river and water exchange with coastal waters. Due to the strong influence of semidiurnal tides, with a tidal amplitude of 3 m and a strong seasonality in river discharge, the Guayas River estuary is not stratified (Murray et al., 1975; Cárdenas, 1995; Cifuentes et al., 1996). Average depth in the upper and lower regions are 8 and 10 m, respectively. The partition of the estuary in three regions sacrificed some realism and precision but allowed us to maintain a certain level of simplicity. This approach is necessary given the complexity of physical and biological interactions in this tropical estuary.

Water flux and salinity in the Guayas River estuary were simulated using the method proposed by Miller

and McPherson (1991). This method incorporates effects of fresh water inflows (advection) and tidal flushing (dispersion) on the transport of water and waterborne constituents through a series of sequential, one-dimensional volume elements (boxes) (Fig. 5). The method uses the approach that over many tidal cycles the tidally averaged 'flow' (Q_g , m³/s) of water from the coast with a salinity of 35 g/kg (in our case the Gulf of Guayaquil) can be treated as a constant at any point in the estuary. Q_g then can be used in a simple mixing equation to predict salinity in the estuary at different river flows and to calculate tidally averaged exchange flows (Q_E) between boxes. This method is better applied in long, narrow estuaries in which most of the fresh water inflow is near the head of the estuary as is the case of the Guayas River estuary.

Salinity data used in the model were collected in a series of cruises along the Guayas river estuary at near monthly intervals from 1989 to 1990 under different regimes of river discharge (Cárdenas, 1995). River discharge data used to calculate Q_g was an average among values obtained in the same period (Navarrete, CEDEGE, personal communication). Q_E , the average exchange flow, was calculated by assuming that fresh water inflow with a salinity of 0.1 g/kg, mixes with the necessary transport of net water (Q_g) from the Gulf of Guayaquil, with an average salinity of 29.2 g/kg (at Jambelli passage, Station P3, Fig. 3). The observed salinity is then calculated using:

$$S_i = \frac{(S_g * Q_g) + (S_r * Q_{ri})}{Q_g + Q_{ri}} \quad (1)$$

where S_i is the observed salinity at a given site during sampling period i , S_g the salinity of the Jambelli passage end member (29.2 g/kg), Q_g the tidally average flow of new Gulf water needed at a given site to produce the observed salinity when mixed with a fresh water of flow Q_{ri} , S_r the salinity of river-end member (0.1 g/kg), and Q_{ri} the average antecedent inflow of fresh water for sampling period i . Q_g is relatively independent of river inflow and was treated as a constant at each sampling site (Miller and McPherson, 1991). We used the same least-square normal equation derived from Eq. (1) as presented in Miller and McPherson (1991) (the numerator in the last term of their Eq. (2) should be Q_{ri}^2 instead of Q_{ri}) to calculate values of Q_g .

Exchange flow (Q_E , m^3/s) between boxes was calculated as follows:

$$Q_{Ei} = \frac{(S_i - S_{i-1})(Q_r + Q_{Ei-1})}{(S_{i+1} + S_i)} \quad (2)$$

where Q_{Ei} is the tidally averaged flow of water due to tidal action needed to maintain a salt balance between box i and $i + 1$. The upstream region of the Guayas river estuary ($i = 1$) S_{i-1} was set to the salinity calculated for the upstream boundary of box 1, and Q_{Ei-1} was set to 0 in the most downstream box ($i = 3$), S_{i+1} was set to the salinity of the downstream boundary of box 3 (Station P3, Fig. 3).

The exchange flows were used to calculate salinity in each box at each DT:

$$C_{Ei} = C_{Si} + \frac{dt}{V_i} [Q_r(C_{Si-1} - C_{Si}) + Q_{Ei-1}(C_{Si-1} - C_{Si}) + Q_E(C_{Si+1} - C_{Si})] \quad (3)$$

where C_{Ei} is the concentration of a conservative constituent (i.e. salt) in the longitudinal center of box i at the end of the current time step, and C_{Si} the concentration of the constituent in the longitudinal center of box i at the start of the current time step (Miller and

McPherson, 1991). The terms included in the brackets calculate the effects due to river inflow of advection on concentration, as well as the effects of tidal circulation and dispersion from the adjacent upstream and downstream boxes (Fig. 3).

Water turnover time is only 11 d as a consequence of high tide amplitude leading to large fluxes of water and nutrients across the boundaries that determine system behavior. Although we did not use hourly time steps to specifically model diurnal inequalities of water levels during tides (Murray et al., 1975), a time step of 1 day allowed us to correctly simulate water fluxes and salinity oscillations. Given the high turnover rate of water in this estuary, hydrological features such as a vertical layers and structure in the water column do not completely develop and therefore are not included in the model.

3.1.2. Exchange of sediment and nitrogen

Equations representing transport of total suspended sediments and total nitrogen through the estuary are similar to those that describe the transport of salt. Seasonal concentrations of suspended sediment and total nitrogen in Guayas River and in offshore waters were based on water quality surveys in the area (Table 1). The model determines the exchange, or net flux, of total nitrogen, and suspended sediments among the upper, middle and lower zones of the estuary. Sediments and nitrogen in the model flows not only among the three estuarine zones, but also between bay waters, shrimp ponds, and the surrounding mangroves within each zone (Fig. 5). Each ha of ponds or mangroves contributes a fixed amount of nitrogen or captures a fixed amount of sediments. Thus, the model simplifies their interaction with the estuary by using exchange coefficients for sediment and nitrogen between either mangroves or shrimp ponds and the estuary. Mangroves remove nitrogen but ponds are a source of this nutrient because of fertilization. However, both mangroves and shrimp ponds remove sediment, thus they have no distinct function in the model (but different magnitude). Sediment accumulates in mangroves contributing to the expansion of this wetland in the estuary; whereas sediment accumulated in shrimp ponds is dredged periodically and accumulates on land (Twilley, 1989). In addition, both nitrogen and sediment are lost in each segment of the estuary. Sedimentation of suspended sediment is estimated at 1%/d of the total amount of sediment in each respective estuarine zone. Loss of nitrogen is linked to both sedimentation and denitrification and we estimated this was 10%/d of nitrogen content in each respective zone of the Guayas River estuary.

Mangroves may be a sink of sediments and nutrients in coastal waters (Lynch et al., 1989; Twilley, 1997a) and they have been recommended for the treatment of

nutrient enriched effluent (Macnae, 1968; Corredor and Morell, 1994), although this function is still poorly understood (Clough et al., 1983). The Guayas River estuary is turbid and extensive mangroves along the shore line trap sediments suspended in the water column during flooding, enriching mangrove soils (Twilley, 1995). Sediment accumulation in the mangroves of the Guayas River estuary averages about $3660 \text{ g m}^{-2} \text{ yr}^{-1}$, or about $10 \text{ g m}^{-2} \text{ d}^{-1}$, and this rate was used in the model. Nitrogen accumulation in the model is based on a rate of $11.3 \text{ g m}^{-2} \text{ yr}^{-1}$, or about $0.03 \text{ g m}^{-2} \text{ d}^{-1}$ (Twilley, 1995). Much of this nitrogen accumulation in mangrove sediments is supported by inorganic nutrients carried by tides from the estuary into the wetland (Rivera-Monroy et al., 1995a) and little of this inorganic nitrogen is apparently lost by denitrification at unenriched concentrations (Rivera-Monroy et al., 1995b; Rivera-Monroy and Twilley, 1996). However, mangroves have a very high potential to process excess nitrogen either by sediment accumulation or denitrification (Corredor and Morell, 1994; Rivera-Monroy et al., in press).

During shrimp pond management, the natural exchange of estuarine water via tides with the intertidal zone has been replaced with diesel pumps that link shrimp ponds to the estuary. Diesel engines are used to pump water daily from the estuary during high tides to a central aqueduct system where water then flows to individual ponds. Exchange rates vary from 3 to 8% of the pond volume per day under semi-extensive mariculture, and may increase to 10–15% under more intense farming practices (Boyd and Fast, 1992). The total volume of water pumped from the Guayas River estuary to shrimp ponds depends on exchange rates (% of pond volume/d) and area of ponds in operation (Twilley, 1989). At a present operation of 92,000 ha of ponds under semi-extensive management (5%/d pumping rate), the volume of water exchanged daily with the estuary is approximately $65 \times 10^6 \text{ m}^3$ (using a mean pond depth of 1.5 m; Twilley, 1989; Twilley et al., 1993). This volume exceeds the discharge of the Guayas River during low flow period. At intensive pond management (10% pumping rate), the same area of ponds would exchange more volume than river discharge during high flow. These types of scenarios underscore the potential significance of water use by shrimp ponds on the pattern of water quality in this estuary.

Supplemental feeding and fertilization methods are required to meet the demand for food at higher stocking densities of post larvae in ponds. A main source of nutrition for shrimp in growout ponds are phytoplankton blooms that result from urea and superphosphates added prior to stocking. Supplemental feeding is carried out towards the end of the growth cycle, usually the last four weeks. Much of the nitrogen and phos-

phate applied to ponds are absorbed by phytoplankton and are thus returned to the estuary in organic form (Rivera-Monroy et al., in press). We used a loading rate of $0.1 \text{ g m}^{-2} \text{ d}^{-1}$ ($365 \text{ kg ha}^{-1} \text{ yr}^{-1}$) of total nitrogen from shrimp ponds operations to the estuary. This value varies among types of operation and includes assumptions that most of the organic nitrogen is exported (Boyd and Fast, 1992). This value needs better calibration.

3.1.3. Scenarios

The incremental impact of replacing mangroves with shrimp ponds was investigated by running different scenarios of land use in the intertidal zone. The model allows ponds to replace mangrove wetlands in the intertidal zone. In run S1, the entire intertidal area is considered mangroves with no ponds (100:0); in run S2 the intertidal area is equally distributed between mangroves and shrimp ponds (50:50); and in run S3, all of the intertidal area is converted to shrimp ponds (0:100). The intertidal area (km^2) for each zone of the estuary used in each simulation run was computed directly from maps of the area (Fig. 5). When 100% of the intertidal zone is in mangroves, the area (km^2) of mangroves in each specific zone is equal to total intertidal area as noted in Fig. 5. If mangroves and ponds are 50% each of the intertidal area, each occupies half the specific intertidal area of each respective zone (run S2). The simulations are compared to the base run where all land is in mangroves and river flow is at 100% of 1989 levels. Each of the intertidal land use ratios (100:0, 50:50, 0:100, % of mangrove:pond) were run at 100, 50 and 10% of base line river flow.

3.2. Simulation results and discussions

Seasonal patterns of salinity were under conditions of 1989 river flow were similar in simulations compared to field measurements. Comparisons of salinity in Estero Salado based on simulation output compared to field measures were used to validate the hydrographic design of the model (data not shown). In simulations for this experiment of intertidal land use, we varied river flow to test the effect of hydrography on environmental quality. Changes in salinity were most sensitive to river flow in the upper estuary, but as expected salinities in all three zones increased with decreased river flow (Fig. 6). Even at only 10% river flow, the highest salinities were 25 g/kg at the mouth of the estuary. The sharpest gradient in salinity was between mid and lower zones during full river discharge, with salinities of 7 and 20 g/kg, respectively. At low river flow (10%), there was less difference in average salinities among the zones.

Mangroves remove suspended sediments in the model, as do shrimp ponds; thus even though one of

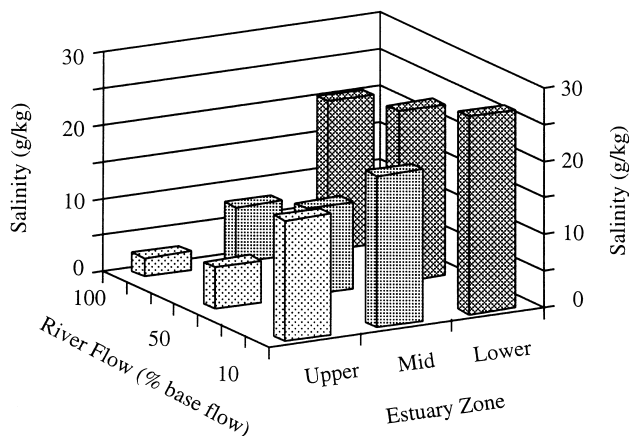


Fig. 6. Simulation results of salinity in three zones of the Guayas River estuary showing the three reductions of river flow.

these ecosystems replaces the other, the function of the intertidal zone remains as a sediment sink. Thus, as mangroves are converted to shrimp ponds, the sediment sink function of the intertidal zone actually increases, because ponds are more efficient at accumulating sediment than are mangroves. Accordingly, the suspended sediment concentrations in the base run of normal river flow and tides with all mangroves in the intertidal zone (100:0) yield suspended concentrations of 375 mg/L in the upper estuary compared to about 300 mg/L when the intertidal zone is all ponds (0:100) (Fig. 7). The other factor that decreases sediment concentrations is the amount of river flow. At 90% reduction in river flow, suspended concentrations are only 5–20% of base run concentrations because sediments have more time to settle out. Again, changes in suspended sediment are most apparent in the upper and middle regions of the estuary with little change in the lower estuary (Fig. 7). Concentrations are most sensitive to magnitude of river flow rather than land use.

Total nitrogen concentrations change with land use depending on the amount of river flow in the estuary (Fig. 8). With all mangroves and no shrimp ponds in each sector of the estuary (100:0), as river flow decreases there is an increase in total nitrogen in the estuary in the upper and middle zones, but concentrations remain low in the lower estuary with all land use patterns. At 90% reduction in river flow, total nitrogen concentrations increase by 5 fold in the upper and middle estuary. In addition, with a reduction in mangroves and increased construction of shrimp ponds to a 50:50 distribution in each region of the estuary, concentrations of total nitrogen increase 30 fold in the upper estuary compared to 10 fold in the middle estuary (50:50 compared to 100, 50 and 10% river flow). Again, there is very little change in the total nitrogen in the lower estuary associated with land use change at reduced river flow. Extreme nitrogen

concentrations are observed in the upper and middle estuary when all the mangroves are removed and replaced with shrimp ponds (0:100), with 60 and 30 fold increases, respectively. Peak concentrations, at nearly 275 μM of total nitrogen are extreme eutrophic concentrations. Note that the middle bay shows about one-third of the increase in the upper bay (Fig. 5). In summary, decreased river flow combined with the conversion of mangroves to shrimp ponds tends to increase nitrogen concentration markedly in the upper and middle bays, with less change in the lower estuary.

There exists an important potential interaction between mangroves and shrimp ponds with respect to total nitrogen and suspended sediment concentrations

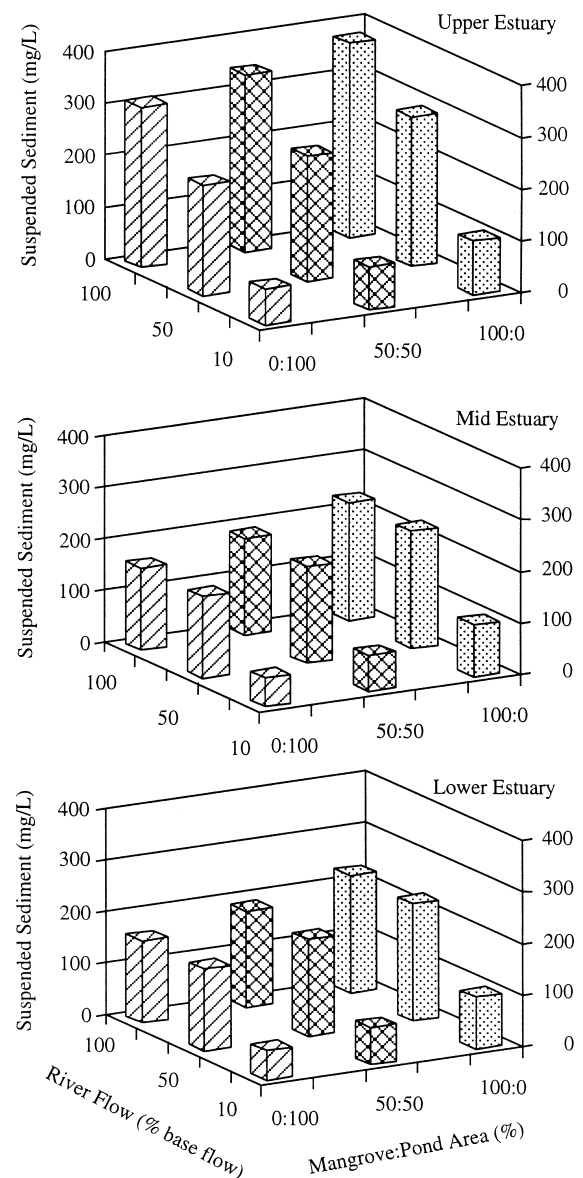


Fig. 7. Simulation results of suspended sediments in three zones of the Guayas River estuary showing the three land use scenarios at three reductions of river flow.

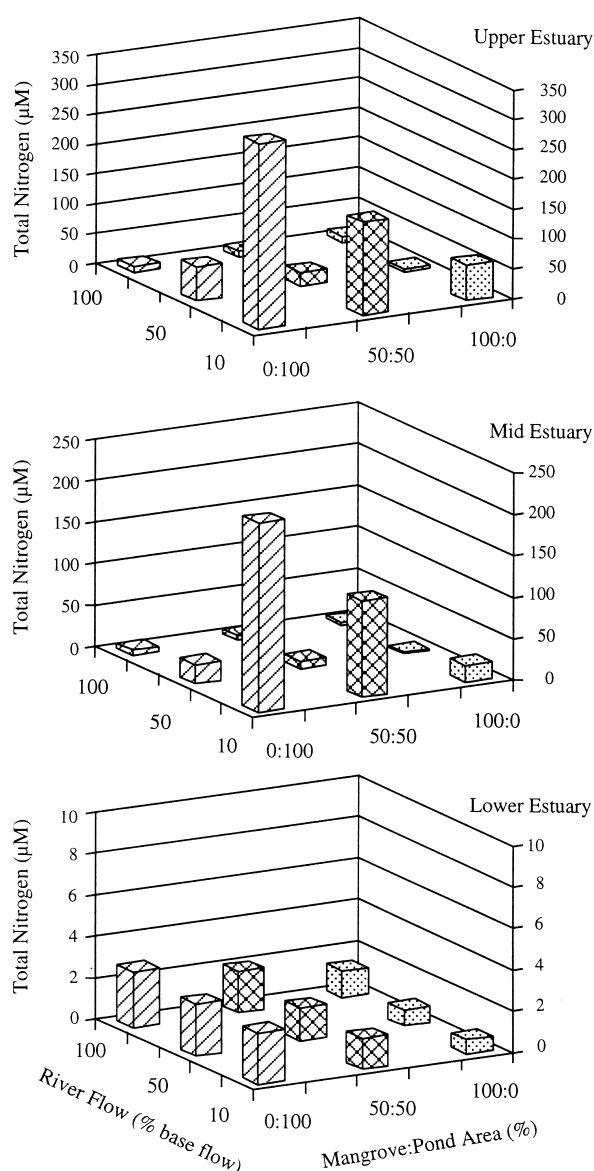


Fig. 8. Simulation results of total nitrogen in three zones of the Guayas River estuary showing the three land use scenarios at three reductions of river flow.

in response to land use in the intertidal zone. A decrease in suspended sediment concentration due to pond construction allows more light to penetrate the water column. This light is an important resource for the growth of phytoplankton in normally turbid river-dominated estuaries. Therefore, the reduction of suspended sediment alone will probably promote phytoplankton growth and the productivity of this pond/estuary ecosystem. But those scenarios that caused a reduction in suspended sediment concentrations also resulted in increased total nitrogen concentrations as ponds enrich estuarine waters and mangroves no longer remove nutrients. Thus, these two land use changes will increase two resources: light and nutri-

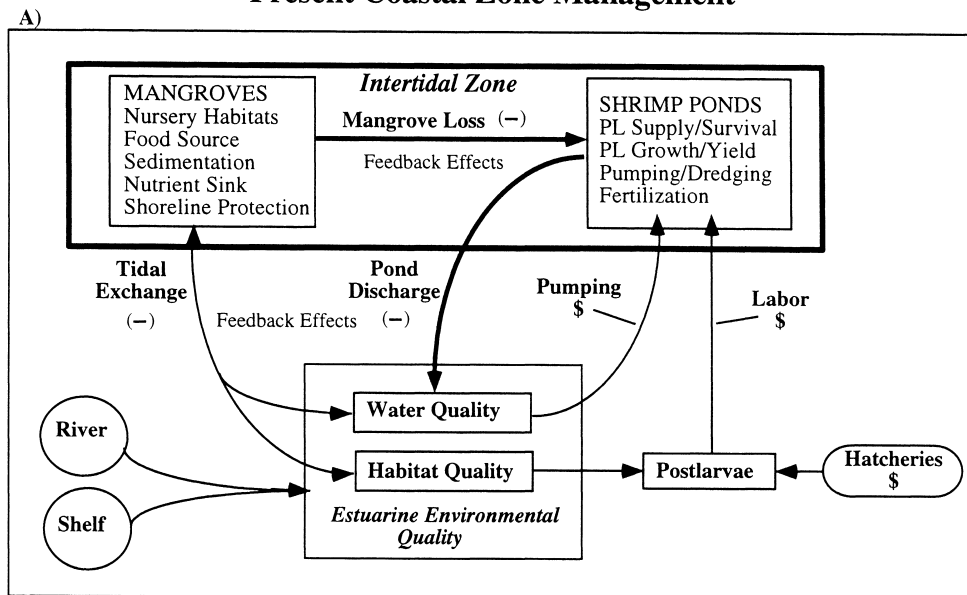
ents. The combination of these two factors should markedly change the productivity and water quality leading to increased potential for the eutrophication of the Guayas River estuary.

Model simulations indicate that shrimp pond industries in the Guayas River estuary have managed to sustain high productivities and profits largely as a consequence of substantial river discharge to this tropical estuary. Seasonally high water input reduces the residence time of nutrients in the estuary, preventing significant eutrophication problems. The government's construction of the Daule-Peripa Dam upstream, however, threatens local economic industries by substantially reducing river flow, perhaps by about one-half. The model indicates that nitrogen concentrations will increase in the upper and middle regions of the estuary to levels that threaten the environmental quality of the Guayas River estuary. However, there will be much less effect on the estuarine resources in the lower zone, where concentrations remain low because of the high exchange with mesotrophic waters in the Gulf of Guayaquil. Suspended sediment concentrations will decrease similarly to various levels across the three estuarine zones when the dam is under full operation. These simulations and preliminary analyses suggest that changes threatening the eutrophication of this estuary will vary according to location along the shoreline. Water quality problems leading to reduced profitability of the shrimp industry will also be spatially selective. Impacts of land use changes in the upland watershed will not be evenly distributed along the Guayas River estuary. In addition, these changes, particularly the reduction of fresh water discharge can potentially reduce the environmental quality of the estuary.

4. Integrated mangrove management

The long term sustainability of shrimp mariculture in Ecuador will require integrative approaches to the management of coastal zone resources. These management considerations not only include the interactions of the shrimp farming industry with estuarine and coastal ocean ecosystems, but also land use activities in the upland watershed and urban centers. For example, the interactions of shrimp farming activities with the Guayas River estuarine ecosystem demonstrate the complex nature of how environmental impacts influence the sustainability of this industry (Figs. 2 and 9). The upper panel of Fig. 9 describes the present strategy in utilizing the coastal resources of Ecuador by the shrimp mariculture industry, including the feedback effect of this enterprise on estuarine environmental quality. Decreases in the environmental quality of estuarine resources affect the productivity of

Present Coastal Zone Management



Integrated Mangrove Management

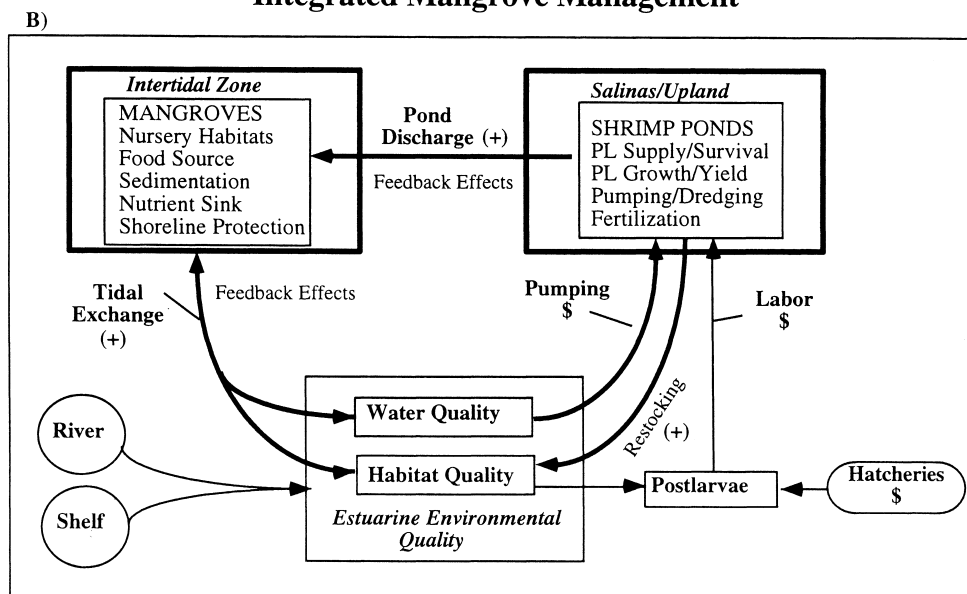


Fig. 9. Comparison of present practices in coastal zone management (A) with recommended strategies for integrated mangrove management (B) based on the ecological processes of mangroves and their function in maintaining the habitat and water quality of the coastal zone; and how these functions support shrimp farm industry.

shrimp ponds by influencing the ability of natural resources to supply post larvae, and by controlling the survival and growth of shrimp in growout ponds. Thus, the secondary productivity of pond ecosystems is constrained by the variety of factors that are linked to water and habitat quality in coastal ecosystems (Figs. 2 and 9).

There are some indications that mangroves can be managed and engineered to remove excessive nutrients

in coastal environments (Nedwell, 1975; Corredor and Morell, 1994; Robertson and Phillips, 1995; Rivera-Monroy et al., in press). This is a particularly important research agenda given the increased eutrophication of coastal waters in the tropics. Mangroves may represent sinks of several primary nutrients used in the fertilization of ponds, particularly phosphates and nitrogen. Mangrove sediments may also have the capacity to absorb some of the BOD associated with

pond effluent high in chlorophyll biomass that may shift the balance of dissolved oxygen in the estuary. Effluent from shrimp ponds could be distributed in nearby mangrove forests for nutrient removal prior to the return of water back to the estuary. The use of mangroves as a nutrient filter of pond effluent would limit the negative feedback of shrimp ponds on the water quality of coastal ecosystems (Fig. 9B). The contribution of excess nutrients from shrimp ponds to mangrove ecosystems would most likely enhance mangrove productivity and potentially enhance the contribution of these forested wetlands to habitat quality of estuaries. This scheme to integrate the natural function of mangroves with the management of shrimp ponds would serve as a means of altering what is presently a negative impact of intensive aquaculture to estuarine ecosystems into a positive feedback (Fig. 9B). The shift from extensive to intensive mariculture may not necessarily impact the estuarine ecosystem if mangroves could be utilized in the operation of these types of ponds.

Integrating mangroves into shrimp pond operations will require more substantial evidence of its cost effectiveness, rather than using recommendations based solely on ecological function. Quantitative estimates of mangrove loss with corresponding increase in pond operations need to be linked to changes in environmental quality of the estuary. In addition, incremental impacts will most likely vary spatially along the longitudinal axis of the estuary, due to variation of water residence time and rates of ecological processes. Thus, site selection of ponds may be a key management option to sustaining both environmental quality of the estuary and minimizing costs of pond operations (Snedaker et al., 1986). Finally, the functional ecology of mangroves, both natural and restored wetlands, need to be compared and integrated with the engineered services of shrimp ponds (Fig. 1).

These simulations have demonstrated the sensitivity of key hydrologic and land use parameters to water quality issues in the Guayas River estuary. Differences in river flow influence the sensitivity of water quality in the estuary to land use of intertidal zone between mangrove conservation and shrimp pond construction. These parameters, in turn, will affect the role that mangroves play in determining industry profits at different zones along the estuary. Accordingly, the model may enable planners and conservationists to determine those estuaries, and particular locations within estuaries, where industry profits may be particularly vulnerable to the loss of mangroves due to the unique physical characteristics of estuaries. One of the initial results of these simulations is that mangroves do not always play an important role in maintaining water quality. Using self-interest as a motivation for maintaining mangroves will work best when self-inter-

est can be demonstrated reasonably to pond owners. The model suggests that self-interest can only be so demonstrated under certain hydrographic conditions of the estuary. Pond management strategies represent behavioral forcing functions that affect the ecology of the entire estuary. And these pond operations are linked to the comprehensive land use alternatives in the entire river basin that influence the environmental quality of the estuary.

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